Enhancing UAV Coverage Path Planning via Deep Reinforcement learning with Intrinsic Curiosity

1st Yinxin Zhou
School of Computer Science
Nanjing University of Posts and Telecommunications
Nanjing, China
b21031704@njupt.edu.cn

2nd Yanxi Chen
School of Computer Science
Nanjing University of Posts and Telecommunications
Nanjing, China
b21030201@njupt.edu.cn

3rd Renyu Wang
School of Electronic Information and Communications
Huazhong University of Science and Technology
Wuhan, China
wry2021@hust.edu.cn

4th Haiping Huang
School of Computer Science
Nanjing University of Posts and Telecommunications
Nanjing, China
hhp@njupt.edu.cn

5th Shenghao Liu
School of Cyber Science and Engineering
Huazhong University of Science and Technology
Wuhan, China
liushenghao@hust.edu.cn

6th Xianjun Deng
School of Cyber Science and Engineering
Huazhong University of Science and Technology
Wuhan, China
dengxj615@hust.edu.cn

2024 IEEE Smart World Congress (SWC) | 979-8-3315-2086-1/24/$31.00 ©2024 IEEE | DOI: 10.1109/SWC62898.2024.00095

***Abstract*—Unmanned Aerial Vehicles (UAVs) are widely used in various fields due to their agility and versatility, but their limited energy supply necessitates efficient Coverage Path Planning (CPP). Traditional CPP methods are effective in structured environments but often fall short in more complex scenarios. This paper proposes a novel approach integrating advanced reinforcement learning techniques to enhance UAV path planning. Our method combines Double Deep Q-Networks (DDQN) with Long Short-Term Memory (LSTM) networks to improve decision-making and temporal information processing capabilities. Additionally, we incorporate an Intrinsic Curiosity Module (ICM) to drive exploration and Prioritized Experience Replay (PER) to focus on high-value experiences, accelerating learning and enhancing performance. The proposed framework demonstrates superior coverage efficiency and adaptability in simulations, achieving higher coverage ratios and reduced redundant paths compared to traditional methods. These innovations provide a robust and flexible solution for autonomous UAV navigation, offering a solid foundation for future research in CPP optimization.**

Keywords—Coverage Path Planning, Reinforcement Learning, Double Deep Q-Networks

I. Introduction

Unmanned Aerial Vehicles (UAVs), due to their agility and versatility, have found widespread applications across diverse fields such as smart farming [1], disaster management [3], wildfire tracking[5] and so on. Their capacity to swiftly navigate and cover areas from above has revolutionized data collection and monitoring practices, enabling precise, real-time insights that are critical in various operations. As UAVs are often powered with limited energy supply, these applications typically require UAVs to perform a *coverage path planning* (CPP) task to ensure that the area of interests can be efficiently inspected before energy runout.

The core of CPP is to plan a path that allows UAVs to traverse all target areas while minimizing energy consumption. Traditional methods such as the boustrophedon[7], Morse decomposition based coverage algorithms[9] and the spiral path coverage algorithm[11] have been proposed to solve the CPP problem. Additionally, various heuristic algorithms, genetic algorithms, etc., have also been widely used to solve the CPP problem. Although these methods have shown excellent path planning efficiency in structured environments, they typically require a substantial amount of prior knowledge about the specific environment, performing poorly when dealing with dynamic changes and complex environments. This challenge is similar to coverage problems encountered in sensor networks, where ensuring comprehensive coverage with limited resources has been extensively studied[15]. These studies highlight the importance of adaptive strategies to maintain coverage in uncertain or dynamic environments[17].

Reinforcement learning (RL) is a family of algorithms that allow an agent to continuously update and optimize its strategy through interaction with the environment, notably without the need for prior knowledge. The development of RL technology has provided a new approach to solving this problem. Hung et al. [15] combined swarm algorithms and Q-learning to address the path planning of multiple UAVs in stochastic environments. With the development of deep neural networks, Mnih et al. [16] utilized neural networks to estimate value functions, introducing the Deep Q Network (DQN) based on value functions. Van Hasselt et al.[17] proposed the Double Deep Q Networks, which better improves the performance of DQN. Building on this, Luis et al. [18] utilized DQN and DDQN methods to train agents for full coverage of lakes, while Apuroop et al.[19] employed Actor-Critic Experience Replay to train convolutional neural networks, aiding bricklaying robots in more efficiently completing cleaning tasks.

Although current reinforcement learning technologies have significantly optimized the efficiency of Coverage Path Planning (CPP), they often fail to effectively utilize the data generated during the training process. This limitation results in a lack of adaptability and flexibility when facing complex and dynamic environments. Additionally, existing techniques frequently overlook the enhancement of intrinsic motivation for UAVs to explore uncharted areas, leading to suboptimal path planning outcomes in such scenarios.

To overcome these challenges, we propose a novel approach that integrates several advanced reinforcement learning techniques to enhance UAV path planning. Our method combines Double Deep Q-Networks (DDQN) with Long Short-Term Memory (LSTM) networks, which improves decision-making and temporal information processing capabilities. This integration allows the UAV to better utilize historical data and adapt to changes in the environment. Furthermore, we incorporate an Intrinsic Curiosity Module (ICM) to drive exploration, significantly

2993-396X/24/$31.00 ©2024 IEEE
DOI 10.1109/SWC62898.2024.00095

Authorized licensed use limited to: Southeast University. Downloaded on January 05,2026 at 04:53:53 UTC from IEEE Xplore. Restrictions apply.

improving the UAV's environmental comprehension and area coverage, especially in uncharted or evolving landscapes. By introducing Prioritized Experience Replay (PER) into our training process, we focus on experiences that offer the most informational value, thereby accelerating the learning process and enhancing performance. Additionally, a sophisticated reward structure is developed to promote efficient exploration and minimize redundant flight paths. This ensures the optimal use of the UAV's limited battery resources for maximum area coverage. Collectively, these innovations address the limitations of current reinforcement learning technologies, providing a robust and flexible framework for autonomous UAV navigation and offering a solid platform for future research.

II. Problem Formulation

A. Environment Modeling

Given the intuitive representation and the simplicity in data handling offered by grid maps, this study selects the grid map model as the foundational framework for map modeling. When employing the grid map model, the resolution of the grid significantly impacts the efficiency of the algorithm's execution. Excessively high resolutions can lead to data redundancy, while too low resolutions may overlook crucial map details. Therefore, this paper determines the size of the grid based on the coverage range of unmanned aerial vehicles (UAVs), ensuring that each grid corresponds precisely to the area covered by a UAV in one pass.

Based on the principles above, this research constructs the task map as a grid model and represents it through a two-dimensional array $G(x,y)$, where x and y represent the horizontal and vertical coordinates on the grid map respectively. Considering that UAVs need to differentiate between flyable areas and no-fly zones when executing full coverage path planning tasks, this study uses a notation of 0 and 1 within the grid map to distinguish these two areas: flyable areas are marked by 0, while no-fly zones are marked by 1. To further detail the UAV's coverage status, additional markers are introduced into $G(x,y)$ to differentiate between already covered grids and newly covered grids by the UAV, thus enabling precise monitoring and analysis of the UAV's coverage behavior.

The UAV must avoid areas designated as no-fly zones to ensure safe and legal operation. These areas are represented on the grid map G as cells with a value of 1. The UAV's path planning algorithm must account for these restrictions, ensuring that no planned path intersects these restricted areas.

The no-fly zone constraint is formulated as:

$$G_m \cap G_n = \varnothing$$

where G_m is the area covered by UAVs during the task, and G_n is the no-fly zone in the grid map.

B. Objective Function

To better evaluate the efficiency of UAV coverage path planning, we use two indicators: Coverage Ratio(CR) and Repeat Ratio(RR).

The Coverage Ratio(CR) measures the proportion of the total flyable area that the UAV successfully covers during its mission, which can be calculated as:

$$CR = \frac{S_c}{S_t}$$

where S_c is the area covered by the UAV during the mission; S_t is the total flyable area.

The Repeat Ratio(RR) assesses the extent to which the UAV covers the same area more than once, which can be calculated as:

$$RR = \frac{S_r}{S_c}$$

where S_r is the area covered by the UAV more than once; S_c is the area covered by the UAV during the mission.

A higher coverage ratio indicates better task effectiveness, whereas a higher repetition ratio indicates lower task efficiency. Thus, the goal of CPP is to maximize the Coverage Rate while minimizing the Repeat Rate within a given constraint, which can be combined through weighted summation. The objective function can be formulated as:

$$Maximize\ F = k_1 \cdot CR - k_2 \cdot RR$$

where k_1 and k_2 measures the weight of CR and RR respectively.

C. Constraints

In the context of UAV coverage path planning, energy consumption and task duration are critical constraints. To simplify the modeling of energy consumption and task duration constraints, we introduce a unified budget constraint based on the number of movement steps the UAV can take. By equating each movement step to a specific amount of energy consumption, we streamline the planning process, allowing for more straightforward and effective optimization of the UAV's path. This unified approach ensures that the UAV operates efficiently within its power limits.

Each movement step directly correlates to an incremental use of the UAV's battery power. Thus, the total number of permissible steps provides a clear and quantifiable metric to gauge the UAV's operational capacity. By capping the total number of steps, we control the UAV's energy usage, ensuring it does not exceed its battery capacity. This prevents the UAV from running out of power mid-mission, which could result in incomplete coverage or loss of the UAV. Limiting the number of steps also indirectly sets a boundary on the mission duration. Since each step takes a finite amount of time, the step limit ensures that the mission can be completed within a reasonable timeframe, aligning with practical operational constraints.

The unified budget constraint can be represented as:

$$\sum_{t=1}^{T} \delta_t \leqslant N$$

where T represents the total time steps; δ_t denotes the movement steps taken at each time step t; N represents the constraint time step.

By integrating this constraint, we ensure that the UAV can effectively plan its path within the given energy and time

Authorized licensed use limited to: Southeast University. Downloaded on January 05,2026 at 04:53:53 UTC from IEEE Xplore. Restrictions apply.

limitations, maximizing coverage while maintaining operational efficiency.

III. Methodology

In this section, we introduce a novel framework for UAV coverage path planning, which integrates Double Deep Q-Networks (DDQN) with Long Short-Term Memory (LSTM) networks, Prioritized Experience Replay (PER), and an Intrinsic Curiosity Module (ICM). This proposed method enhances the UAV's path planning capabilities by combining advanced decision-making, temporal information processing, and incentivized exploration. The overall architecture of our model is illustrated in Fig. 1, which highlights the neural network structure along with the embedded ICM module.

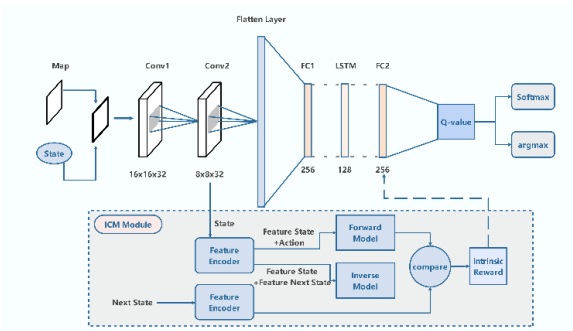


Fig. 1. Model structure

A. Neural Network Model

The architecture of our model is designed to process both spatial and temporal information effectively. Thus, the UAV's position is converted into a 2D one-hot representation, encoding the occupied cell within the entire grid. This encoding, combined with the grid map representation, forms the input to the network's convolutional layers.

Two convolutional layers are employed to extract spatial features from the input state. The convolutional layers are padded to ensure that their output shape remains consistent with the input shape. All layers use zero-padding, with the rectified linear unit (ReLU) chosen as the activation function. The output from the convolutional layers is flattened into a one-dimensional feature vector and passed through a fully connected layer, transforming the high-dimensional convolutional feature maps into a lower-dimensional form suitable for processing by the LSTM layer.

The output from the convolutional layers is then flattened into a one-dimensional feature vector and passed to an LSTM layer with a hidden state dimension of 128. The LSTM layer encodes temporal information, learning from the sequence of past actions and observations, which is crucial for capturing the dynamics of the environment over time. After the LSTM layer, a fully connected layer maps the extracted features to Q-values corresponding to each possible action in the UAV's action space. This layer provides a quantitative evaluation of action outcomes, allowing the UAV to make informed decisions.

The training procedure involves the use of Prioritized Experience Replay (PER) to focus on experiences that offer the most informational value, accelerating the learning process and enhancing performance. The DDQN mitigates the overestimation bias of traditional DQN by using separate networks for action selection and evaluation. This separation ensures a more reliable estimation of action values, reducing the positive bias that the original DQN is prone to.

B. Reinforcement Learning Module

To ensure both stability and accuracy in the learning process, we utilize the Double Deep Q-Network (DDQN) algorithm, which improves upon traditional Q-learning. Reinforcement learning aims to determine the optimal actions to take in various states to maximize rewards, based on a Markov Decision Process (MDP). An MDP consists of a state space, an action space, state transition probabilities, and a reward function. The goal is to learn a policy, which is a strategy for choosing actions based on the current state, to maximize cumulative rewards over time.

Traditional Q-learning estimates the value (Q-value) of taking a particular action a in a specific state s, establishing a two-dimensional Q-table to record the optimal action and reward for each state. Q-learning learns the optimal strategy by iteratively updating the Q-table, with the formula as follows:

$$Q(s,a) = Q(s,a) + \alpha[r + \gamma \cdot \max(Q(s',a')) - Q(s,a)]$$

where α represents the learning rate; γ is the discount factor that determines the current value of future rewards; s' indicates the new state after taking action a; r signifies the immediate reward received from taking action and $\max(Q(s',a'))$ is the maximum Q value among all possible actions.

Deep Q-Networks (DQN) address these issues by using deep neural networks to approximate the Q-function. The network, represented by parameters θ inputs state information and outputs Q-values for each possible action, thus managing high-dimensional state spaces more effectively. DQN introduces two significant innovations: experience replay and the use of a target network. Experiences (s_t, a_t, r_t, s_{t+1}) are stored in a memory pool D and sampled randomly during training to break the correlation between consecutive experiences and stabilize learning. Additionally, two networks are used: an evaluation network and a target network. The target network provides stable target Q-values for training and is updated periodically to follow the evaluation network's parameters. The DQN update rule is:

$$y_t = r_{t+1} + \gamma \max_a Q(s_{t+1}, a; \theta^-)$$

where θ^- represents the parameters of the target network.

Despite these improvements, DQN can overestimate Q-values, leading to suboptimal policies. DDQN mitigates this issue by separating the action selection and action evaluation processes between two networks, thereby reducing the overestimation bias. The DDQN update rule is:

$$y_t = r_{t+1} + \gamma Q(s_{t+1}, arg\max_a Q(s_{t+1}, a; \theta^-); \theta)$$

$$y_t = r_{t+1} + \gamma Q(s_{t+1}, arg\max_a Q(s_{t+1}, a; \theta^-); \theta)$$

In this setup, the evaluation network selects the action, while the target network evaluates its Q-value. The target

Authorized licensed use limited to: Southeast University. Downloaded on January 05,2026 at 04:53:53 UTC from IEEE Xplore. Restrictions apply.

network is periodically updated with the parameters from the evaluation network to ensure stable learning.

To further enhance learning efficiency, we implement Prioritized Experience Replay (PER). PER prioritizes experiences with higher Temporal Difference (TD) errors, as these are more critical for learning. A sum-tree data structure is used to efficiently manage and sample these priorities. The loss function is adjusted with importance-sampling weights to maintain stability and reduce bias:

$$L(\theta) = E_{(s,a,r,s')\sim U(D)}[(r + Q(s', arg\max_{a'} Q(s', a'; \theta); \theta^{-}) - Q(s, a; \theta))^{2}]$$

By focusing on the most significant experiences, PER accelerates the convergence of the learning process. Overall, DDQN, combined with PER, effectively reduces overestimation and improves the accuracy and efficiency of policy learning. This robust framework is well-suited for training deep neural networks in complex environments, optimizing decision-making processes across various applications.

C. Exploration Module

In the realm of reinforcement learning, the external motivation of an agent is to maximize the rewards received from interactions with the environment. However, in scenarios where external rewards are sparse, relying solely on external motivation is insufficient for effective learning. Curiosity-driven approaches represent a class of methods that leverage intrinsic rewards to guide agents in exploring the environment. Through efficient exploration, these methods can accelerate the acquisition of external rewards[20], while also reducing the certainty of environmental rewards and state transitions, thus balancing exploration with exploitation[21].

To drive exploration, we utilize the Intrinsic Curiosity Model(ICM), which provides intrinsic rewards based on prediction errors. The ICM consists of two models: an inverse model and a forward model. The inverse model functions by predicting the action taken by the agent given its current state and the subsequent state. It utilizes the current state s_t and the next state s_{t+1} to predict the action $\hat{a}_t$. The discrepancy between the predicted action $\hat{a}_t$ and the actual action a_t generates an inverse prediction error, which is used to train an encoder. This encoder captures features that are directly relevant to the agent's actions, effectively filtering out environmental noise and focusing on elements that the agent can control.

Simultaneously, the forward model predicts the feature vector of the next state $\hat{\phi}(s_{t+1})$ based on the current state's feature vector $\phi(s_t)$ and the action a_t. Thus, the intrinsic reward is derived from the prediction error, calculated as the difference between the predicted next state features $\hat{\phi}(s_{t+1})$ and the actual next state features $\phi(s_{t+1})$. This prediction error serves as a measure of novelty, encouraging the UAV to explore unvisited or less familiar regions of the environment. By doing so, the UAV can discover new areas and enhance its overall coverage efficiency.

By incorporating the ICM, the UAV is driven by curiosity to explore parts of the environment that it has not yet encountered, even in the absence of external rewards. This method helps in effectively covering the entire area, avoiding redundant paths, and improving the efficiency of the coverage mission. Besides, the ICM enhances the UAV's exploration capabilities, making it a robust tool for environments with sparse or delayed rewards, ensuring comprehensive and efficient coverage of the target area.

D. Reward Function

External rewards are derived directly from the environment's feedback to the agent's specific actions, reflecting the direct value of the agent's behavior in achieving specific goals. These rewards serve as the immediate incentive for the UAV to learn and update its actions during the reinforcement learning process. Therefore, to effectively encourage the UAV to explore efficiently and cover comprehensively while reducing the repetition rate, the reward function in this paper is designed as follows:

The reward function considers different scenarios based on the UAV's state after taking an action:

- Effective Coverage: When the UAV enters an uncovered grid, it effectively covers a new area. In this case, a positive reward is assigned.
- Revisiting Covered Areas: If the UAV's next move involves revisiting an already covered grid, a lower positive reward or no reward is assigned to discourage repetition.
- Proximity to New Areas: When the UAV enters a grid adjacent to an uncovered area, indicating it is close to covering a new area, a moderate positive reward is given.
- Entering No-Fly Zones or Out of Bounds: If the UAV's action leads to entering a no-fly zone or exiting the map boundary, a significant negative reward is assigned to prevent such actions.
- Task Completion: When the UAV successfully covers all areas, completing the task, a substantial positive reward is given.

This reward structure aims to balance the exploration and exploitation strategies of the UAV, encouraging it to cover new areas efficiently while avoiding unnecessary repetitions and collisions.

Based on the Exploration Module, the reward function in our framework is defined as the sum of external rewards and intrinsic curiosity rewards, fostering a dual-reward system that balances immediate goal achievement with exploratory behavior. Thus, the reward function can be calculated as follow:

$$R = r + \lambda R_{ICM}$$

where λ is the weight that measures the importance of the two reward factors.

Through this, we can foster a learning mechanism that not only seeks to maximize immediate external rewards but also engages in exploratory actions that are intrinsically motivated by curiosity. This dual-reward system facilitates a more robust and comprehensive learning process, enabling agents to effectively navigate and learn from their environment, especially in contexts where external rewards are not readily available or are sparse.

The structure of the reward function is shown in Fig. 2.

Authorized licensed use limited to: Southeast University. Downloaded on January 05,2026 at 04:53:53 UTC from IEEE Xplore. Restrictions apply.

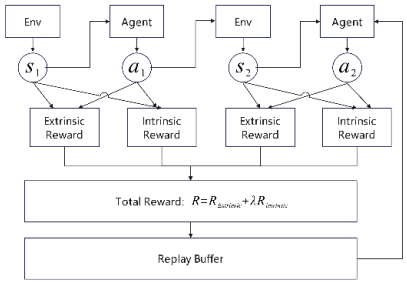


Fig. 2. Structure of the reward function

This figure depicts the interaction between the environment and the UAV, where the UAV receives a state from the environment and takes an action. The environment provides extrinsic rewards based on this action, while intrinsic rewards are generated by the UAV's exploration and curiosity-driven behaviors. This process repeats with subsequent states and actions, leading to the total reward, which is stored in the replay buffer for learning updates. This design ensures that the UAV effectively balances goal-oriented behavior with exploratory learning.

IV. Simulation Result and Discussion

A. Simulation Setup

In the simulation process, the above environmental parameters and other network model parameters are initialized first. To evaluate the effectiveness of the proposed approach, simulations were conducted on a 25x25 grid map with no-fly zones. The grid map was designed to represent a realistic environment with both flyable areas and no-fly zones. The UAVs were tested under various budget constraints of 200, 350, 500, 650, 800, 950, and 1050 steps. Four different algorithms were compared: the proposed DDQN with LSTM and ICM (referred to as ICDDQN), DDQN with Prioritized Experience Replay (DDQN-PER), standard DDQN, and traditional DQN. The performance of each algorithm was measured based on the Coverage Ratio (CR) and the Repeat Ratio (RR). The setting of hyperparameters for model training are listed in TABLE Ⅰ.

TABLE I. Hyperparameters for Model Training

Parameter	Value	Description
M	256	Batch Size
D	10000	Replay buffer size
γ	0.98	Discount factor
ξ_d	3000	ξ decay
LR	0.0006	Learning rate
τ	0.005	Target network update factor
β	0.1	Temperature parameter
N	10000	Maximum number of training episodes

Before each training starts, the position of the UAV appears randomly on the grid map, and it is ensured that the position of the UAV is not the position of the obstacle.

B. Evaluation

Through the previous analysis, we designed maps to simulate three distinct cases: one with many no-fly zones, one with few no-fly zones, and one on a smaller grid map. We conducted simulation experiments in each of these three cases to evaluate the performance of the algorithms under different conditions.

First, in the case of few no-fly zones, the performance of the four algorithms under different budget constraints is illustrated in Fig. 3.

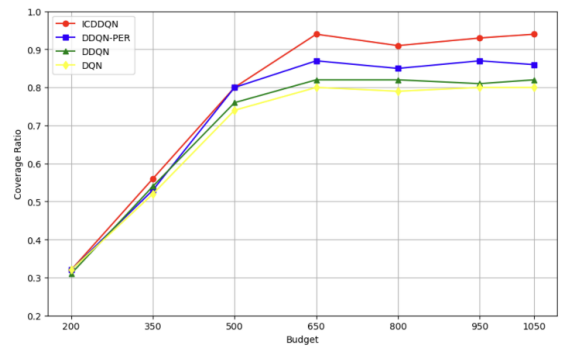


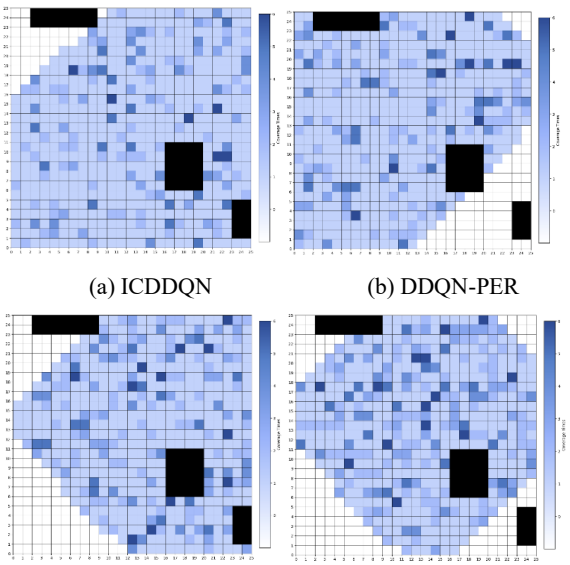
Fig. 3. Coverage ratio with varying budget of the four algorithms

As illustrated in Fig. 3, all algorithms demonstrate an increasing coverage ratio as the budget increases from 200 to 1050 steps, indicating that a higher budget allows UAVs to cover more area. The ICDDQN algorithm consistently outperforms the others across all budget levels, with particularly strong performance at higher budgets where it maintains near-optimal coverage. DDQN-PER also shows strong performance, surpassing both standard DDQN and DQN, benefiting from its prioritized experience replay mechanism. Standard DDQN demonstrates moderate performance, consistently outperforming DQN but falling behind the enhanced versions. DQN exhibits the lowest coverage ratio, especially at higher budgets, highlighting its relative inefficiency.

At lower budgets (200 and 350 steps), the differences between the algorithms are less pronounced, suggesting that the benefits of advanced techniques like LSTM, ICM, and PER are not fully realized under limited steps. However, even at these lower budgets, ICDDQN begins to demonstrate its superiority, indicating its robustness across different scenarios. At higher budgets (800, 950, and 1050 steps), the gap between ICDDQN and the other algorithms becomes more significant. ICDDQN maintains a coverage ratio close to 1.0, showcasing its efficiency in utilizing the available budget. DDQN-PER also maintains strong performance but experiences a slight drop compared to ICDDQN, while both DDQN and DQN exhibit a performance plateau as the budget increases.

In addition to the overall performance evaluation, we conducted a detailed visualization of the coverage patterns for the four algorithms at a budget of 650 steps, as shown in Fig. 4.

Authorized licensed use limited to: Southeast University. Downloaded on January 05,2026 at 04:53:53 UTC from IEEE Xplore. Restrictions apply.



(a) ICDDQN (b) DDQN-PER

(c) DDQN (d) DQN

Fig. 4. Coverage plot for four different algorithms (white: flyable grids, black: no-fly zones, shades of blue: covered areas with varying coverage rates, deeper blue indicating higher coverage)

The coverage ratios for ICDDQN, DDQN-PER, DDQN, and DQN at this budget level are 94%, 87%, 82%, and 80%, respectively. These results reveal distinct differences in the coverage patterns among the algorithms. The ICDDQN algorithm, achieving the highest coverage rate of 94%, demonstrates a thorough and efficient coverage of the grid map, with minimal uncovered areas and a strategic approach to avoiding no-fly zones. DDQN-PER, with a coverage rate of 87%, also shows strong performance but with slightly more uncovered regions compared to ICDDQN. The standard DDQN, at 82% coverage, exhibits a more scattered pattern, covering less area efficiently compared to the enhanced versions. DQN, with the lowest coverage rate of 80%, shows significant inefficiency in maximizing the area coverage.

To further analyze the algorithms' performance over time, we plotted the coverage rate as a function of episodes in the same complex environment with many no-fly zones, as shown in Fig. 5.

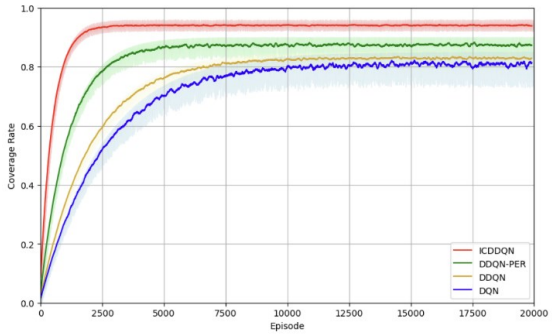


Fig. 5. Coverage rate trend over episodes for different algorithms

Fig. 5: Coverage rate trend over episodes for different algorithms in a complex environment with many no-fly zones

The above figure illustrates the coverage rate trend across episodes for the four algorithms under a scenario with few no-fly zones with a budget of 650 steps. As depicted, the ICDDQN quickly reaches a high coverage rate and maintains it across episodes, consistently outperforming the other algorithms. DDQN-PER also demonstrates robust performance, although slightly less efficient than ICDDQN. Standard DDQN and DQN exhibit slower convergence and lower overall coverage rates, with DQN facing the most difficulty in adapting to the environment. This result highlights the significance of advanced techniques like ICDDQN and DDQN-PER in enhancing UAV coverage efficiency.

In the scenario with many no-fly zones, the performance trends of the four algorithms are similar to those observed with fewer no-fly zones, as illustrated in Fig. 6.

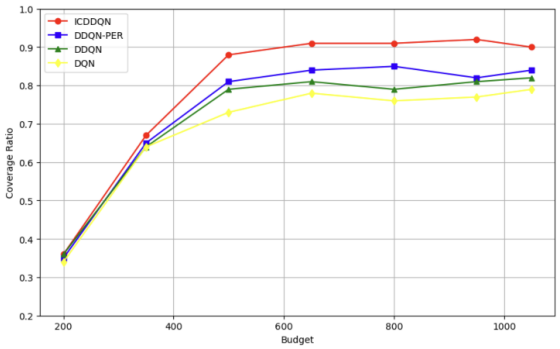
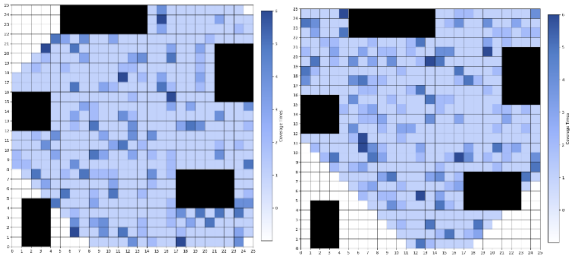


Fig. 6. Coverage plot for four different algorithms (white: flyable grids, black: no-fly zones, shades of blue: covered areas with varying coverage rates, deeper blue indicating higher coverage)

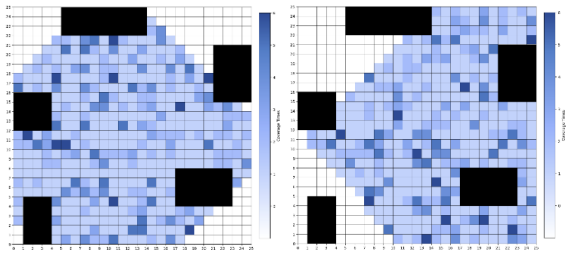
Through the above figure, it can be seen that in the case of many no-fly zones, the performance of the four algorithms and the trend of change is similar to the case of fewer no-fly zones. However, it can be clearly seen that in the case of more no-fly zones, the ICDDQN algorithm relatively shows a greater advantage and better accomplishes the coverage task.

Similarly, we conducted a detailed visualization of the coverage patterns for the four algorithms at the budget of 650 steps, as shown in Fig. 7.



(a) ICDDQN (b) DDQN-PER

Authorized licensed use limited to: Southeast University. Downloaded on January 05,2026 at 04:53:53 UTC from IEEE Xplore. Restrictions apply.



(c) DDQN (d) DQN

Fig. 7. Coverage plot for four different algorithms (white: flyable grids, black: no-fly zones, shades of blue: covered areas with varying coverage rates, deeper blue indicating higher coverage)

The above figures demonstrate the ability of each algorithm to adapt to a more complex environment. The coverage ratios for ICDDQN, DDQN-PER, DDQN, and DQN at this budget level are 91%, 85%, 81%, and 78%, respectively. The ICDDQN algorithm maintains its superior performance, effectively navigating the increased no-fly zones and achieving a high coverage rate. DDQN-PER continues to perform well but shows a slight decrease in efficiency compared to ICDDQN. The standard DDQN and DQN algorithms exhibit more significant performance degradation, with DQN particularly struggling to achieve effective coverage in the presence of more no-fly zones.

To analyze the performance over time in this more challenging scenario, we plotted the coverage rate trend across episodes for the four algorithms, as shown in Fig. 8.

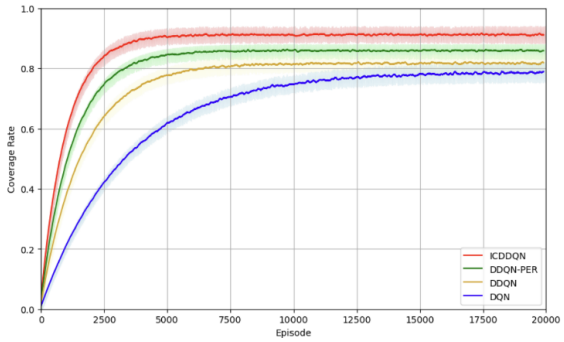
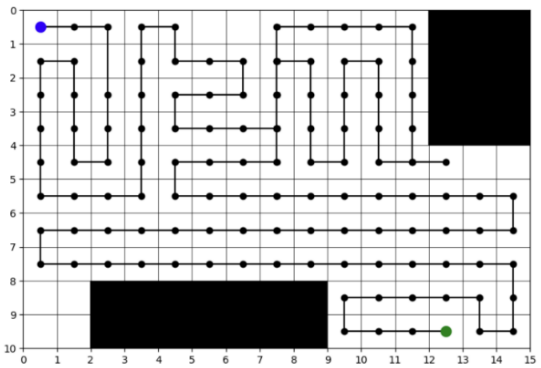


Fig. 8. Coverage rate trend over episodes for different algorithms in a complex environment with many no-fly zones

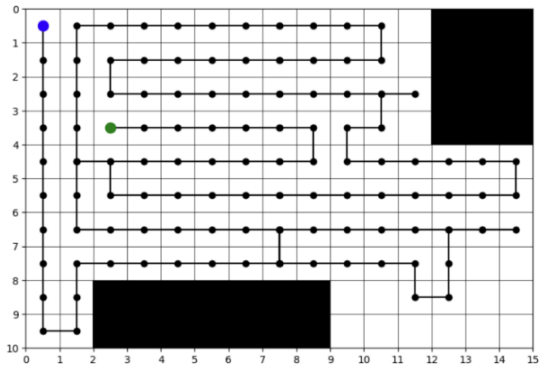
Fig.8 illustrates the coverage rate trend across episodes for the four algorithms under a scenario with a budget of 650 steps. As depicted, the ICDDQN still quickly achieves and maintains a high coverage rate, consistently outperforming the other algorithms. DDQN-PER also exhibits robust performance, though slightly less efficient than ICDDQN. Meanwhile, standard DDQN and DQN show slower convergence and lower overall coverage rates, reinforcing the importance of advanced techniques like ICDDQN and DDQN-PER in improving UAV coverage efficiency in challenging scenarios.

In this case, ICDDQN continues to outperform the other algorithms across various budget levels, maintaining a high coverage ratio even in a more challenging environment. DDQN-PER also shows robust performance, benefiting from the prioritized replay mechanism. The performance gap between the enhanced algorithms (ICDDQN and DDQN-PER) and the standard algorithms (DDQN and DQN) becomes more pronounced in the increased obstacle scenario, highlighting the importance of advanced techniques in improving UAV coverage efficiency.

To further validate the effectiveness of the ICDDQN and DDQN-PER algorithms, we conducted additional experiments on a smaller grid map(10x15). This smaller map presents a different set of challenges, with its reduced size and potentially different obstacle configurations, which help to assess the adaptability and efficiency of the algorithms in diverse environments. The coverage patterns for these algorithms on the 10x15 grid map are illustrated in Fig. 9, where the blue point is the start point and the green point is the end point.



(a) ICDDQN (Coverage ratio: 94%)



(b) DDQN-PER (Coverage ratio: 87%)

Fig. 9. Coverage plot for ICDDQN and DDQN-PER on a 10x15 grid map

The proposed ICDDQN achieved a high coverage ratio of 94%, effectively navigating and covering the smaller grid map with minimal uncovered areas. The algorithm's path shows a strategic and efficient approach, successfully avoiding no-fly zones and maximizing area coverage. DDQN-PER also performed well, with a coverage ratio of 87%, although it leaves slightly more uncovered regions compared to ICDDQN. This suggests that while DDQN-PER benefits from prioritized experience replay, further optimization in efficiency is possible.

The experimental results clearly reinforce the superiority of the ICDDQN algorithm in achieving efficient coverage across various grid sizes and obstacle scenarios. The higher coverage ratio ,of ICDDQN, combined with its strategic path planning, highlights its advanced capabilities in dynamic and

Authorized licensed use limited to: Southeast University. Downloaded on January 05,2026 at 04:53:53 UTC from IEEE Xplore. Restrictions apply.

constrained environments. These results underscore the practical applicability of ICDDQN, especially in real-world scenarios where UAVs must navigate and cover areas of varying sizes with different obstacle distributions. The consistent performance of ICDDQN across different environments suggests its robustness and reliability. The insights gained from these experiments on smaller grid maps can inform future improvements and optimizations in UAV coverage algorithms. The performance gap between ICDDQN and DDQN-PER, although smaller, indicates that further enhancements in learning mechanisms and obstacle navigation strategies can yield even better results.

V. CONCLUSION

In this research, we introduced an innovative method for UAV full-coverage path planning by integrating the advanced decision-making capabilities of Double Deep Q-Networks (DDQN) with the sequential data processing power of Long Short-Term Memory (LSTM) networks. This approach is further enhanced with Prioritized Experience Replay (PER) and an Intrinsic Curiosity Module (ICM), ensuring efficient learning and exploration. The DDQN framework provides robust action selection, while LSTM's ability to utilize past experiences helps navigate complex environments by recognizing patterns over time. PER optimizes the learning process by focusing on more informative experiences, and ICM drives exploration by rewarding the UAV for investigating new areas. This comprehensive integration significantly improves operational efficiency, enabling the UAV to cover areas thoroughly while avoiding unnecessary revisits.

Our simulation results demonstrated that the proposed ICDDQN algorithm consistently outperforms traditional methods, achieving higher coverage ratios and reducing redundant paths across various scenarios, including environments with many no-fly zones, few no-fly zones, and smaller grid maps. The higher coverage ratio of ICDDQN, combined with its strategic path planning, highlights its advanced capabilities in dynamic and constrained environments. These results underscore the practical applicability of ICDDQN, especially in real-world scenarios where UAVs must navigate and cover areas of varying sizes with different obstacle distributions.

Future work will focus on empirical validation of our model in real-world scenarios, incorporating environmental variables such as weather conditions and UAV operational constraints like battery life to refine the model's applicability for autonomous UAV missions. Additionally, exploring further enhancements in learning mechanisms and obstacle navigation strategies can yield even better results. This future work aims to bridge the gap between theoretical robustness and practical applicability, enhancing the utility of autonomous UAVs in critical applications requiring extensive and meticulous area exploration, such as environmental monitoring, agricultural surveillance, and search and rescue operations.

REFERENCES

[1] Lottes, P.; Khanna, R.; Pfeifer, J.; Siegwart, R.; Stachniss, C. UAV-Based Crop and Weed Classification for Smart Farming. In Proceedings of the 2017 IEEE International Conference on Robotics and Automation (ICRA), Singapore, 29 May–3 June 2017; pp. 3024–3031.

[2] E. Galceran and M. Carreras, “A survey on coverage path planning for robotics,” Robotics and Autonomous Systems, vol. 61, no. 12, pp. 1258–1276, 2013.

[3] Erdelj, M.; Natalizio, E.; Chowdhury, K.R.; Akyildiz, I.F. Help from the sky: Leveraging UAVs for disaster management. IEEE Pervasive Comput. 2017, 16, 24–32.

[4] T. Cabreira, L. Brisolara, and P. R Ferreira, 'Survey on coverage path planning with unmanned aerial vehicles,' Drones, vol. 3, no. 1, 2019.

[5] Pham, H.X.; La, H.M.; Feil-Seifer, D.; Deans, M. A Distributed Control Framework for A Team of Unmanned Aerial. In Proceedings of the 2017 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), Vancouver, BC, Canada, 24–28 September 2017; pp. 6648–6653.

[6] D. Gimesy, 'Drones and thermal imaging: saving koalas injured in the bushfires - [news],' The Guardian, 10 Feb 2020.

[7] H. Choset and P. Pignon, “Coverage path planning: The boustrophedon cellular decomposition,” in Field and service robotics. Springer, 1998, pp. 203–209.

[8] E. M. Arkin, S. P. Fekete, and J. S. Mitchell, 'Approximation algorithms for lawn mowing and milling,' Computational Geometry, vol. 17, no. 1-2, pp. 25–50, 2000.

[9] E. U. Acar, H. Choset, A. A. Rizzi, P. N. Atkar, and D. Hull, “Morse decompositions for coverage tasks,” The International Journal of Robotics Research, vol. 21, no. 4, pp. 331–344, 2002.

[10] Y. Zeng, Q. Wu, and R. Zhang, 'Accessing from the sky: A tutorial on UAV communications for 5G and beyond,' Proceedings of the IEEE, vol. 107, no. 12, pp. 2327–2375, 2019.

[11] E. Gonzalez, O. Alvarez, Y. Diaz, C. Parra, and C. Bustacara, “BSA: A complete coverage algorithm,” in Proc. IEEE International Conference on Robotics and Automation (ICRA), Apr. 2005, pp. 2040–2044.

[12] Wang, Bang. "Coverage problems in sensor networks: A survey." ACM Computing Surveys (CSUR) 43, no. 4 (2011): 1-53.

[13] Wang, Bang. Coverage control in sensor networks. Springer Science & Business Media, 2010.

[14] Wang, Bang, Kee Chaing Chua, Vikram Srinivasan, and Wei Wang. "Information coverage in randomly deployed wireless sensor networks." IEEE transactions on Wireless Communications 6, no. 8 (2007): 2994-3004.

[15] Hung S M, Givigi S N. A Q-Learning Approach to Flocking With UAVs in a Stochastic Environment[J].IEEE Transactions on Cybernetics, 2016:186-197.DOI:10.1109/TCYB.2015.2509646.

[16] MNIH V，KAVUKCUOGLU K，SILVER D，et al. Hum-an-level control through deep reinforcement learning［J］. Nature，2015，518（7）：529-533.

[17] VAN HASSELT H，GUEZ A，SILVER D. Deep re-inforcement learning with double Q-learning［EB/OL. arXiv:1509.06461,2015.

[18] Luis, Samuel Yanes, Daniel Gutiérrez Reina, and Sergio L. Toral Marín. "A deep reinforcement learning approach for the patrolling problem of water resources through autonomous surface vehicles: The ypacarai lake case." *IEEE Access* 8 (2020): 204076-204093.

[19] Apuroop, Koppaka Ganesh Sai, Anh Vu Le, Mohan Rajesh Elara, and Bing J. Sheu. "Reinforcement learning-based complete area coverage path planning for a modified hTrihex robot." *Sensors* 21, no. 4 (2021): 1067.

[20] ACHIAM J，SASTRY S. Surprise-based intrinsic motivation for deep reinforcement learning［EB/OL］.（2017-03-06）［2021-02-21］. https：//arxiv. org/pdf/1703. 01732. pdf.

[21] SCHMIDHUBER J. Formal theory of creativity，fun，and intrinsic motivation（1990-2010）［J］. IEEE Transactions on Autonomous Mental Development，2010，2（3）：230-247.

Authorized licensed use limited to: Southeast University. Downloaded on January 05,2026 at 04:53:53 UTC from IEEE Xplore. Restrictions apply.